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Groundwater potential mapping using machine learning models for Northeastern Karbi Anglong district, Assam, India

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***Abstract*—Groundwater is an important source of freshwater for mankind that has the unique capability of recharging itself naturally. Unfortunately, decades of over-exploitation has overwhelmed its ability to replenish, so much so that it has recorded unprecedented levels of decline bordering on depletion that is slowly becoming irreversible. In order to reverse the caused damage and prevent further worsening of the situation, it is important to understand the current status and strategize accordingly. The study was undertaken for identifying best model among a variety of machine learning models for groundwater potential mapping in the Karbi Anglong district, Assam, India. Seven groundwater influencing factors of altitude, slope, aspect, land use, normalized difference vegetation index, rainfall and proximity to rivers were chosen for the study. k-nearest neighbours, linear regression, logistic regression, naïve bayes, linear discriminant analysis, quadratic discriminant analysis, decision trees, random forest, support vector machine, gradient boosted trees, and artificial neural networks are the models that were employed in the study and were compared for finding the most accurate model for delineating groundwater potential in the region. The study found that random forest model achieved the highest accuracy of 91.67% and thus concluded it to be the model best suited for such a study and recommends it for similar other geospatial studies.**

Keywords— Machine learning; Groundwater potential mapping; Random forest; Geospatial analysis; Remote sensing; Geographic information system

I. INTRODUCTION

Water is essential to human existence and hence, it is an established fact that our planet's habitable environment owes itself to the presence of this nectar. While it is well known that 71% of the earth's surface is covered by water, however, it is the meager 1.69% of this entire volume of water on earth, which is found in groundwater reservoirs that comprise the only accessible fresh water source that can truly quench the mankind's thirst. Through various eras of evolution, nature has silently witnessed our growing reliance on this precious resource. The growing demands of a developing economy have since transitioned our modest usage of these resources to a complete blasphemous over-exploitation and are leaving it in complete shambles. It is, hence, a need of the hour to take cognizance of the damage that has been incurred and make drastic amends before the resource completely gives out under the weighty expectations of the new world and takes away the thread tying our existence and survival to the planet. In order to design, plan and finally implement these amends it is a pre-requisite to quantify the current state of these resources and arrive at the right way in going forward with such attempts. It is essential to identify the regions with high susceptibility of groundwater depletion and delineate them from regions that have decent levels of potential to yield groundwater that can continue to supply for the daily demands without being in the risk of running dry and have a high chance of recoverability through natural recharge. These activities are encompassed in a groundwater potential mapping study that tends to employ certain reliable techniques to get the aforementioned results in the form of a groundwater potential zonation map that delivers the outlined outcomes for a particular region under study. These quantified results produced with adequate accuracy could assist in laying down strategies, policies and plans for city planners, agricultural experts, urban estate developers, farmers, cultivators, government bodies, private agents and others for a sustainable and smart deployment of these resources for their respective jobs and services.

II. PREVIOUS STUDIES

Traditionally, groundwater potential mapping studies around the world have applied techniques based on statistics and simple probabilistic sciences. These included trivial linear relations between the dependent and independent factor only and did not have the ability to capture complex phenomena inherent in nature studies. While these methodologies delivered decent accuracies, however, the field has enormous potential for enhancements. A few of the traditional methods comprised of frequency ratio [1], [2], Shannon's entropy [3], [4], analytical hierarchy process [2], etc. Mankind has come a long way with the development of science and technologies such as remote sensing, geographic information system and spatial analysis. Coupled with the evolution of human's analysis skills through artificial intelligence, machine learning, data mining, data analytics has opened up avenues that never could have been thought of before. The culmination of machine learning and geographic information sciences has brought about a completely new perspective of the planet and has unearthed various latent resources both literally and

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figuratively. Literally, since such geospatial studies have helped identify resources suitable for a feasible extraction, such as groundwater [5]–[10], minerals [11]–[13], etc. and figuratively, as there have also been studies that have given significant insights into earth's various geophenomena such as floods [14], landslides [15], etc. It is not surprising, that a lot of studies have come up for groundwater potential mappings applying various machine learning models such as random forest [5]–[8], [10], support vector machines [10], neural networks [16], etc. The objective of the current study was to investigate and compare the performance of various machine learning models for groundwater potential mapping all undertaken in the same region with the same set of attributes and find the best performing predictor.

III. DATA AND STUDY AREA

The region for the study is located in the North-eastern part of the Karbi Anglong district as depicted in Fig. 1. The area under study spans over the Diphu subdivision of Karbi Anglong district in Assam. The district is situated on the south central part of the North-eastern Indian state of Assam. The study area is predominately hilly in terrain and is inhabited by the native tribal communities. The study region spans over an area of around 7319 sq. km and is located between the longitudes of 92°47′44″ E and 93°53′49″ E and the latitudes of 25°31′55″ N and 26°36′53″ N. The ground water in the district is considered suitable for drinking purposes, except for high iron and fluoride contamination beyond permissible limits in certain pockets with the fluoride contamination going as high as 16 parts per million. The groundwater is majorly employed for drinking purposes, however a significant increase in groundwater withdrawal is being observed for meeting the irrigation needs. Groundwater potential mapping requires a directory of locations known to be yielding groundwater in significant yields and the values for the set of factors known to be contributing to the aforementioned yield. The directory forms the inventory part of the dataset, which is augmented with the ground water conditioning factors; both are described in the following sections.

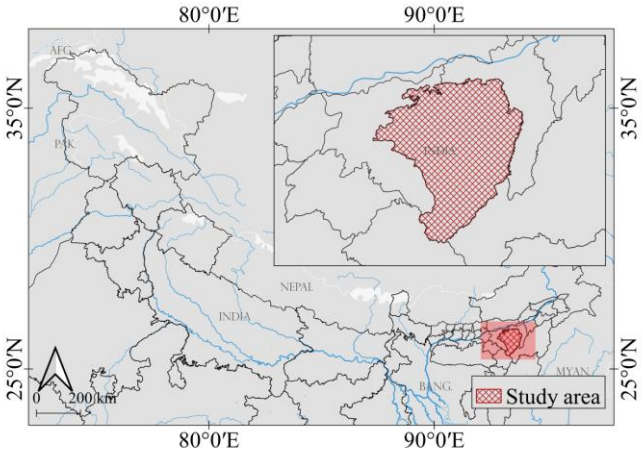


Fig. 1. Location of the study area

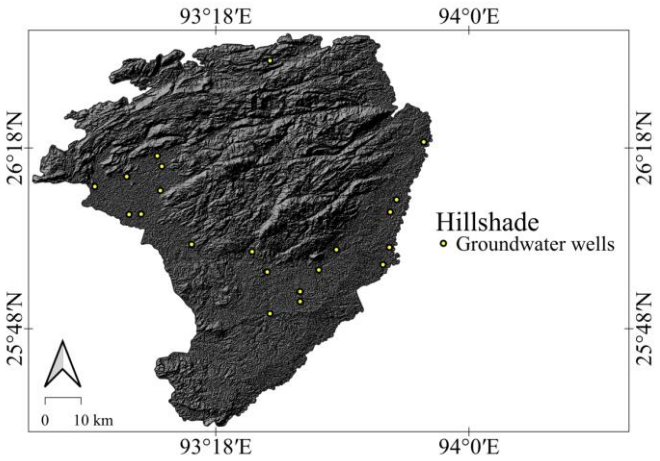


Fig. 2. Locations of groundwater wells in the study region.

A. Inventory

The locations for the 21 of the ground water wells were obtained as part of the groundwater well inventory [17] as depicted in Fig 2. 70 % of these points, i.e. 15 of the locations were used for training the machine learning models, while the remaining 30% of the inventory that accounts for the remaining 6 locations were used for testing the models on the basis of various performance measures such as accuracy, sensitivity, specificity.

B. Conditioning factors

The ground water conditioning factors are a region's characteristics that have a significant impact on the region's ground water yielding potential. The 7 conditioning factors that have been considered in this study are namely altitude, slope angle, slope aspect, land use and land cover, normalized difference vegetation index, rainfall and proximity to rivers depicted in Fig. 3(a-g). A brief description of the aforementioned influencing factors, with their data source for acquisition along with other specific traits characteristic of the region and their significance in impacting the region's ground water yielding potential is as follows. **Altitude** refers to the elevation above the earth's surface. It is depicted in Fig. 3(a) and it was produced from the 2.5 m resolution Digital Elevation Model obtained from Cartosat-1,CartoDEM Version-3 R1[18]. **Slope** is another significant factor influencing a region's groundwater potential. It was also produced from the aforementioned DEM. It is shown in Fig 3(b). Slope accounts for the gradient of the terrain and affects the volume of surface water that infiltrates into underground reservoirs and volume that runs off upland. **Aspect** describes the orientation of the slope of the terrain and has been employed as an influencing factor in a variety of studies [9]. The slope aspect map for the study region was again derived from the region's DEM and is depicted in Fig. 3(c). **Land use Land cover (LULC)** represents the activity/exercise a landscape is invested for. It is established fact that the rigor of the human involvement in land's activities has a direct bearing on the groundwater status underneath the space.

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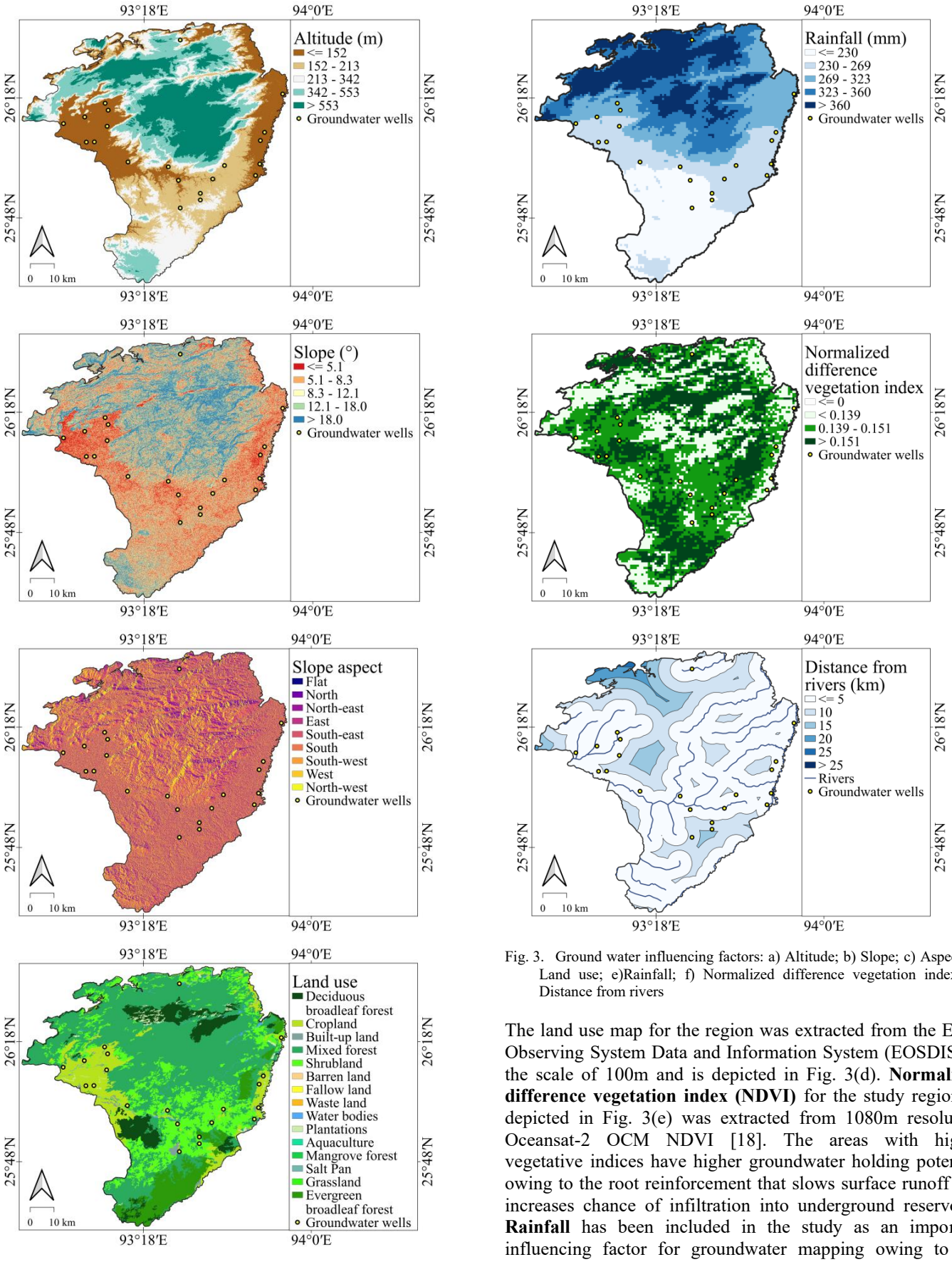


Fig. 3. Ground water influencing factors: a) Altitude; b) Slope; c) Aspect; d) Land use; e)Rainfall; f) Normalized difference vegetation index; g) Distance from rivers

The land use map for the region was extracted from the Earth Observing System Data and Information System (EOSDIS) at the scale of 100m and is depicted in Fig. 3(d). **Normalized difference vegetation index (NDVI)** for the study region as depicted in Fig. 3(e) was extracted from 1080m resolution Oceansat-2 OCM NDVI [18]. The areas with higher vegetative indices have higher groundwater holding potential owing to the root reinforcement that slows surface runoff and increases chance of infiltration into underground reservoirs. **Rainfall** has been included in the study as an important influencing factor for groundwater mapping owing to the

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impact a region's climate conditions have on the state of the groundwater table. It indirectly affects the region's surface water reserves, and the quantity of surplus water to infiltrate after significant volume being lost to runoffs and evaporation. The thematic for the region representing the precipitation in the region was obtained from Global Climate Data at resolution of 1000m (Fig. 3(f)). **Distance from rivers** has a direct bearing on the water table of the region and is depicted in Fig. 3(g). The thematic was obtained and extracted from DIVA GIS. All the factors were transformed into a grid spatial database of 30 x 30m size and the grid for the study region was constructed by 3685 columns and 4004 rows (8,132,720 pixels; 7319 sq. km.).

IV. Methodology

Fig. 4 depicts the methodology adopted in the current study. The steps executed to implement the groundwater potential mapping in the study region are explained as follows. Firstly, a dataset was produced by means of an overlay operation of the inventory thematic over the thematic maps for the ground water influencing factors for the region. The inventory comprised of point coordinates for 21 groundwater well locations, while the conditioning factors selected for the current study comprised of 7 attributes identifying the region's hydro-geo-morphological characteristics, namely elevation, land use, rainfall, normalized difference vegetation index, slope, aspect, and distance from rivers. Since groundwater potential modeling being a binary classification problem, so to avoid bias, 21 non groundwater well locations along with their respective values of the conditioning factors were appended to the dataset. Once compiled the dataset was then partitioned into 2 portions calibrated at 70% and 30% of the original data set.

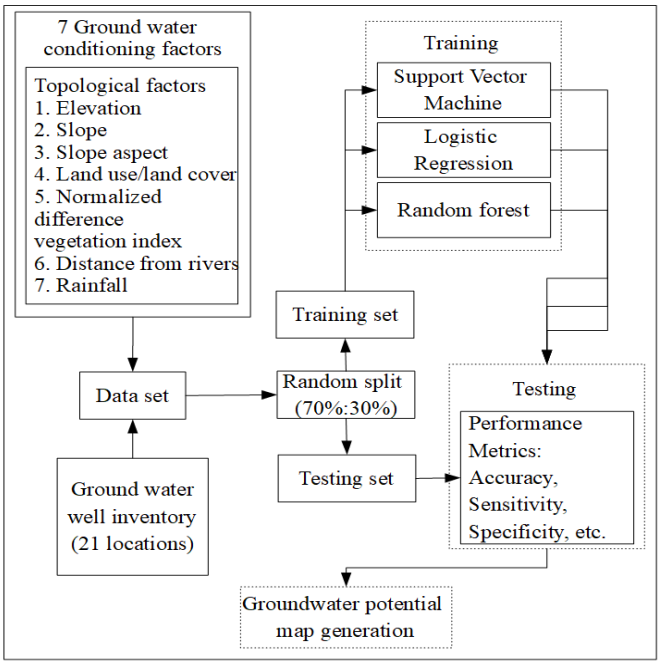


Fig. 4. Methodology employed for ground water potential mapping

The former and the larger portion of the dataset comprising of 70% of the dataset was used for training the various machine learning models, while the latter portion comprising of the remaining 30% records of the original dataset were employed for validating and thus comparing the models on the basis of various performance measures such as sensitivity, specificity, accuracy etc. Upon validation, the model with the best predictive performance was determined on the basis of aforementioned statistical measures and on identifying such model; it was used to generate the groundwater potential index for the entire study region. Next, the computed groundwater potential indices for the entire study region were classified into the categories of very low, low, moderate, high and very high and were used to generate the delineated groundwater potential map for the region. The map was analyzed for area coverage in each delineated zone as per the categorized class across the study region.

V. Random Forest (RF)

The machine learning model of random forest is based on decision trees [7]. In actuality it comprises of an ensemble of decision trees that are built on various bootstrap subsets of the original data set. In each of the trees in the ensemble, for every node the split variable is decided from a subset of all variables. Thus the deciding parameters for the random forest predictor are a) number of trees in the ensemble and b) number of variables in the subset of variables for deciding the split variable at the time of node development for growing the depth of the tree[5], [6], [8]. Once the ensemble of decided number of trees has been developed, the final prediction for an unseen data record is achieved by accumulating the predictions from each of the trees in the ensemble for the data record. This is achieved by traversing each tree using the data record and arriving at the prediction and recording it. Once all the trees have been traversed, the predictions achieved and recorded, the final prediction for the data record from the random forest is said to be obtained by taking a majority vote from all the recorded predictions [19], [20].

VI. Results and Discussion

As can be observed from Table I, Random Forest (RF) model outperformed the rest of the machine learning models that were employed for the groundwater potential mapping in the current study area. The remaining models applied for the task are, namely Linear Regression (LinR), Logistic Regression (LogR), Support Vector Machine (SVM), Artificial Neural Network (ANN), Decision Tree (DT), Gradient Boosted Tree (GBT), Naïve Bayes (NB), Linear Discriminant Analysis (LDA), k-Nearest Neighbors (kNN) and Quadratic Discriminant Analysis (QDA). While, RF achieved an accuracy of 91.67%, the next top three predictors in terms of accuracy after RF are LinR, LogR and SVM each displaying an accuracy of 83.33%. It can also be deduced from the results that LDA, kNN, and QDA are the least accurate classifiers in terms of accuracy achieving the respective values of 58.33%, 50.0% and 50.0%.

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TABLE I. MODEL PERFORMANCE AND THEIR SUBSEQUENT COMPARISON

	TP	*TN*	*FP*	*FN*	*PPV (%)*	*NPV (%)*	*SEN (%)*	*SPE (%)*	*ACC (%)*
RF	6	5	1	0	85.7	100	100	83.3	91.6
LinR	6	4	2	0	75	100	100	66.6	83.3
LogR	5	5	1	1	83.3	83.3	83.3	83.3	83.3
SVM	6	4	2	0	75	100	100	66.6	83.3
ANN	4	5	1	2	80	71.4	66.6	83.3	75
DT	4	5	1	2	80	71.4	66.6	83.3	75
GBT	4	4	2	2	66.6	66.6	66.6	66.6	66.6
NB	3	4	2	3	60	57.1	50	66.6	58.3
LDA	6	1	5	0	54.5	100	100	16.6	58.3
kNN	2	4	2	4	50	50	33.3	66.6	50
QDA	0	6	0	6	-	50	0	100	50

Note: Formulas for metrics used in Table I are given as ***TP****=True positive;* ***TN****=True negative;* ***FP****=False positive;* ***FN****=False negative;* ***PPV*** *= TP / (TP + FP);* ***NPV*** *= TN / (TN + FN);* ***SEN*** *= TP / (TP + FN);* ***SPE*** *= TN / (TN + FP);* ***ACC*** *= (TP + TN) / (TP + TN + FP + FN)*

RF displayed a perfect sensitivity of 100% and a decent specificity of 83.33%, along with good measures of positive predictive value and negative predictive value at 85.71% and 100%, respectively, thus could be unanimously established as an excellent choice for a predictor in the current study. From the aforementioned observations, it can be deduced with significant confidence that RF performs the best among all the machine learning models deployed for groundwater potential mapping in the area under study. For further validation of this hypothesis the receiver operating characteristic (ROC) curve was plotted for the RF as shown in Fig. 5, with the false positive rate being depicted on the x-axis and the true positive rate on the y-axis. An important measure for quantifying this ability is derived from the ROC as area under the curve (AUC). A perfect classifier would have an AUC of 1, while AUC values ranging between 0.5 to 0.6 depicts a weak classifier, between 0.6 to 0.7 a fair classifier, between 0.7 to 0.8 a good predictor and between 0.8 to 0.9 a very good model and a near perfect model is for AUC > 0.9. Fig. 5 depicts that RF achieved an AUC of 0.9722 which validates RF as a nearly perfect model. Hence, RF was used to generate the groundwater potential map for the study region as shown in Fig. 6. The map delineated the region into 5 zones based on the range of their respective potential and corresponding class that it lies in, as summarized in Table II.

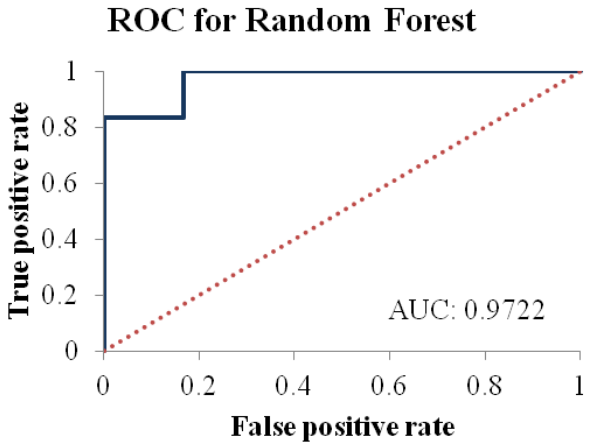


Fig. 5. Receiver operating characteristic curve for groundwater potential mapping using Random Forest model

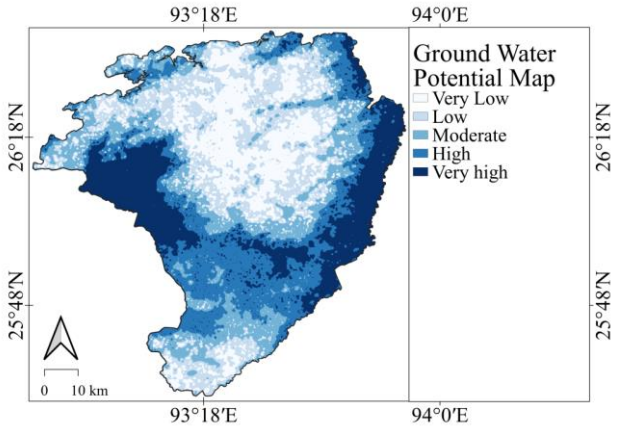


Fig. 6. Groundwater potential delineation map using Random Forest model

This was achieved through the popular classification method of quantile [21]. As per the results depicted in Table II, it can be deduced that around 19.51% of the area has a very high potential to yield ground water while an equally large portion i.e. 21.10% of the area has high groundwater potential.

TABLE II. GROUNDWATER POTENTIAL ZONE DISTRIBUTION USING RANDOM FOREST MODEL

Class	*Groundwater potential range*	*Area (sq.km)*	*% of area*
Very low	0 - 0.1711	1435.45	19.61
Low	0.1711 – 0.2136	1496.21	20.44
Moderate	0.2136 – 0.3545	1414.22	19.32
High	0.3545 – 0.5553	1544.52	21.10
Very high	0.5553 – 1.0	1428.64	19.51

VII. CONCLUSION

Machine learning has established itself as a leading mechanism for modeling data and is finding applications in a variety of fields. Geo-spatial mapping is one such area where machine learning is beginning to find its roots and deliver significant outcomes as it had been producing in other scientific areas. GIS based groundwater potential mapping is a novel area for this kind of data analytics and hence provides a challenge for machine learning predictors. In this study, we evaluated the performance of various machine learning models for groundwater potential mapping in the Northeastern Karbi-Anglong district. From the results of the validation phase, we found that random forest model achieved the highest accuracy of 91.67% and was then used to plot its ROC curve, which revealed an AUC of 0.9722. The model was also employed to delineate the study region into zones based on their potential to yield groundwater which revealed that approximately 19.51% of the region has very high groundwater potential. The outcomes from the study could aid in the long term planning and management of groundwater in the region.

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